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MULTIVARIATE SIMULTANEOUS GENERALIZED ARCH

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This paper presents theoretical results on the formulation and estimation of multivariate generalized ARCH models within simultaneous equations systems. A new parameterization of the multivariate ARCH process is proposed, and equivalence relations are discussed for the various ARCH parameterizations. Constraints sufficient to guarantee the positive definiteness of the conditional covariance matrices are developed, and necessary and sufficient conditions for covariance stationarity are presented. Identification and maximum likelihood estimation of the parameters in the simultaneous equations context are also covered.

1. INTRODUCTION

Although economists have long been interested in the analysis of behavior under uncertainty, econometricians have only recently begun developing an analytical framework to deal with uncertainty. A central feature of this framework is the modeling of second and possibly higher moments, as well. One of the most prominent tools used to model the second moments is due to Engle (1982). Engle (1982) suggested that these unobservable second moments could be modeled by specifying a functional form for the conditional variance and modeling the first and second moments jointly, giving what is called in the literature the Autoregressive Conditional Heteroskedasticity (ARCH) model. Of course, many different functional forms are possible, but Engle's (1982) suggestion that the conditional variances depend on elements in the information set in an autoregressive manner has become perhaps the most common. This linear ARCH model was generalized by Boller-

This paper began as a synthesis of at least three UCSD Ph.D. dissertations on various aspects of multivariate ARCH modeling by Yoshi Baba, Dennis Kraft, and Ken Kroner. In fact, an early version of this paper was written by Baba, Engle, Kraft, and Kroner, which led to the acronym BEKK, which is used in this paper for the new parameterization presented. In the interests of continuity, we maintain the acronym BEKK even though two of the authors have gone on to other pursuits. In addition to Yoshi Baba and Dennis Kraft, we also thank Tim Bollerslev, Doc Ghose, Bruce Hansen, Jan Magnus, Dan Nelson, Ron Oaxaca, Hal White, Jeff Wooldridge, and two thorough referees for fruitful discussion and comments, but, of course, we must accept full responsibility for all errors ourselves. Ken Kroner acknowledges financial support from the Karl Eller Center at the University of Arizona, and Robert Engle from NSF SES 89-10273.

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slev (1986) in a manner analogous to the extension from AR to ARMA models in traditional times series by allowing past conditional variances to appear in the current conditional variance equation. The resulting model is called Generalized ARCH, or GARCH. These models have been applied extensively in the literature (see, e.g., the survey by Bollerslev, Chou, and Kroner, 1992).

Further extensions to multivariate models, which are usually analogous to the extension from ARMA to vector ARMA models, appear often in the literature, though usually without theoretical discussion (see, e.g., Bollerslev, Engle, and Wooldridge, 1988; Engel and Rodrigues, 1989; Engle, Granger, and Kraft, 1984; Kaminsky and Peruga, 1990; Kroner and Claessens, 1991; Kroner and Sultan, 1993; McCurdy and Morgan, 1991; among several others). Multivariate ARCH models allow the variances and covariances to depend on the information set in a vector ARMA manner and are particularly useful in multivariate financial models (such as the CAPM or dynamic hedging models), which require the modeling of both variances and covariances. But while most applications of multivariate ARCH have been to financial modeling, several potential applications also exist in macroeconomics and in other areas of economics. For example, it is often conjectured that employment decreases with price level uncertainty (Friedman, 1977). A hypothesis like this could be tested with the following two-equation model:

$$\begin{aligned} Y &= f(P, X, \sigma_p) \\ P &= g(Y, X), \end{aligned} \tag{1.1}$$

where Y is employment, P is the price level, X are exogenous variables, and σ_p is the ARCH measure of price uncertainty. A significantly negative coefficient on σ_p would provide support for the hypothesis.

The purpose of this paper is to examine the theoretical properties of multivariate generalized ARCH models and to apply these models to systems of simultaneous equations where the second moments of the random variables may be regressors. The paper is organized as follows: Section 2 presents the models, discusses the positive definiteness of the covariance matrix, and examines covariance stationarity of the model; Section 3 analyzes the multivariate GARCH-in-mean model in a simultaneous equations framework; Section 4 discusses estimation of the model; and Section 5 gives some concluding remarks.

2. THE MODELS

2.1. Univariate GARCH

The parameterization of the conditional variance used by Engle (1982) to model the unobservable second moments allows the conditional variance to depend on the elements of the information set in an autoregressive man-

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ner. Letting $\Im_{t-1}$ be the sigma field generated by the past values of ϵ_t, with σ_t^2 measurable with respect to $\Im_{t-1}$, the linear univariate ARCH model can be written as

$$\epsilon_t | \Im_{t-1} \sim N(0, \sigma_t^2)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \cdots + \alpha_p \epsilon_{t-p}^2.$$

This model is called ARCH of order p, or ARCH(p).

Bollerslev (1986) generalized the ARCH process by allowing past conditional variances to appear in the current conditional variance equation. His variance equation becomes

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \cdots + \alpha_{t-q}^2 + \beta_1 \sigma_{t-1}^2 + \cdots + \beta_p \sigma_{t-p}^2.$$

This process is called generalized ARCH of order (p,q), or GARCH(p,q). The simple GARCH(1,1) model often provides a parsimonious description of the data (see, e.g., Bollerslev, 1986; McCurdy and Morgan, 1988).

2.2. Multivariate GARCH

The extension from a univariate GARCH model to an n-variate model requires allowing the conditional variance–covariance matrix of the n-dimensional zero mean random variables ϵ_t to depend on elements of the information set. Letting H_t be measurable with respect to $\Im_{t-1}$, the multivariate GARCH model can be written as

$$\epsilon_t | \Im_{t-1} \sim N(0, H_t).$$

The parameterization for H_t as a function of the information set $\Im_{t-1}$ chosen here allows each element of H_t to depend on q lagged values of the squares and cross-products of ϵ_t, as well as p lagged values of the elements of H_t, and a $J \times 1$ vector of weakly exogenous variables (as defined by Engle, Hendry, and Richard, 1983), x_t. So the elements of the covariance matrix follow a vector ARMAX process in squares and cross-products of the residuals. We will assume x_t contains only current and lagged exogenous variables. Defining

$$h_t = \text{vec}\, H_t$$

$$\tilde{x}_t = \text{vec}(x_t x_t')$$

$$\eta_t = \text{vec}(\epsilon_t \epsilon_t'),$$

where vec(·) is the vector operator that stacks the columns of the matrix, a parameterization can be written:

$$h_t = C_0 + C_1 \tilde{x}_t + A_1 \eta_{t-1} + \cdots + A_q \eta_{t-q} + G_1 h_{t-1} + \cdots + G_p h_{t-p},$$

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where C_0 is a $n^2 \times 1$ parameter vector, C_1 is a $n^2 \times J^2$ parameter matrix, and A_i and G_i are $n^2 \times n^2$ parameter matrices. In matrix notation, this becomes

$$h_t = [C_0 \vdots C_1 \vdots A_1 \vdots \ldots \vdots A_q \vdots G_1 \vdots \ldots \vdots G_p] \begin{bmatrix} 1 \\ \tilde{x}_t \\ \eta_{t-1} \\ \vdots \\ h_{t-p} \end{bmatrix}$$

$$= Fz_t$$

$$= (z_t' \otimes I)\operatorname{vec} F$$

$$= Z_t \alpha, \tag{2.1}$$

where

$$z_t' = (1, \tilde{x}_t', \eta_{t-1}', \ldots, \eta_{t-q}', h_{t-1}', \ldots, h_{t-p}')$$

$$F = [C_0 \vdots C_1 \vdots A_1 \vdots \ldots \vdots A_q \vdots G_1 \vdots \ldots \vdots G_p]$$

$$\alpha = \operatorname{vec} F$$

and

$$Z_t = (z_t' \otimes I).$$

Equations (2.1) define a parameterization that we will call the *vec representation*.

To illustrate, consider a simple two-equation GARCH(1,1) *vec* model without exogenous influences. Model (2.1) becomes

$$h_t = \begin{bmatrix} h_{11,t} \\ h_{12,t} \\ h_{22,t} \end{bmatrix}$$

$$= \begin{bmatrix} c_{01} \\ c_{02} \\ c_{03} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} \epsilon_{1,t-1}^2 \\ \epsilon_{1,t-1}\epsilon_{2,t-1} \\ \epsilon_{2,t-1}^2 \end{bmatrix} + \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \begin{bmatrix} h_{11,t-1} \\ h_{12,t-1} \\ h_{22,t-1} \end{bmatrix}.$$

Notice that we have omitted the equation for $h_{21,t}$ and have given no coefficient to $\epsilon_{2,t-1}\epsilon_{1,t-1}$ or $h_{21,t-1}$, as these are clearly redundant, leaving nine free parameters in each of the A_1 and G_1 matrices. Similar redundancies appear in the general n-variate GARCH(1,1) *vec* model. In particular, all the covariance equations appear twice (i.e., there is an equation for $h_{ij,t}$ as well as for $h_{ji,t}$) and all the off-diagonal terms appear twice within each equation (i.e., both of the terms $\epsilon_{i,t-1}\epsilon_{j,t-1}$ and $\epsilon_{j,t-1}\epsilon_{i,t-1}$ and both of the terms $h_{ij,t-1}$

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and $h_{ji,t-1}$ appear in each equation). The redundant terms can be eliminated without affecting the model, leaving a total of $((n(n+1))/2)^2$ unique parameters in each of the A_i and G_i matrices. In a direct formulation of (2.1), there appears to be n^4 parameters in each matrix, but many of these are superfluous.

For empirical implementation, it is desirable to restrict further this parameterization. A natural restriction that was first used in the ARCH context by Engle, Granger, and Kraft (1984) and in the GARCH context by Bollerslev, Engle, and Wooldridge (1988) is the *diagonal representation*, in which each element of the covariance matrix, $h_{jk,t}$, depends only on past values of itself and past values of $\epsilon_{j,t}\epsilon_{k,t}$. That is, variances depend solely on past own squared residuals, and covariances depend solely on past own cross-products of residuals. This seems an intuitively plausible restriction because information about variances is usually revealed in squared residuals, and if the variances are evolving slowly, then past squared residuals should be able to forecast future variances. A similar argument can be made for covariances. In the *vec* model, a *diagonal* representation is obtained if the matrices A_i and G_i are assumed to be diagonal.

To illustrate in the bivariate case, the *diagonal* model is simply

$$
\begin{aligned}
h_t &= \begin{bmatrix} h_{11,t} \\ h_{12,t} \\ h_{22,t} \end{bmatrix} \\
&= \begin{bmatrix} c_{01} \\ c_{02} \\ c_{03} \end{bmatrix} + \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \begin{bmatrix} \epsilon_{1,t-1}^2 \\ \epsilon_{1,t-1}\epsilon_{2,t-1} \\ \epsilon_{2,t-1}^2 \end{bmatrix} + \begin{bmatrix} g_{11} & 0 & 0 \\ 0 & g_{22} & 0 \\ 0 & 0 & g_{33} \end{bmatrix} \begin{bmatrix} h_{11,t-1} \\ h_{12,t-1} \\ h_{22,t-1} \end{bmatrix}
\end{aligned}
$$

or

$$h_{11,t} = c_{01} + a_{11}\epsilon_{1,t-1}^2 + g_{11}h_{11,t-1}$$

$$h_{12,t} = c_{02} + a_{22}\epsilon_{1,t-1}\epsilon_{2,t-1} + g_{22}h_{12,t-1}$$

$$h_{22,t} = c_{03} + a_{33}\epsilon_{2,t-1}^2 + g_{22}h_{22,t-1}.$$

In the bivariate model illustrated here, there are three free parameters in each of the A_1 and G_1 matrices, and in the general n-variate *diagonal* model there are $((n(n+1))/2)$ free parameters in each matrix.

For any parameterization to be sensible, we require that H_t be positive definite for all values of ϵ_t and x_t in the sample space. In the *vec* representation, and even in the *diagonal* representation, this restriction can be difficult to check, let alone impose during estimation. We now propose a new parameterization that easily imposes these restrictions and that eliminates very few if any interesting models allowed by the *vec* representation.

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Consider the following model:

$$H_t = C_0^{*\prime} C_0^* + \sum_{k=1}^{K} C_{1k}^{*\prime} x_t x_t' C_{1k}^* + \sum_{k=1}^{K} \sum_{i=1}^{q} A_{ik}^{*\prime} \epsilon_{t-i} \epsilon_{t-i}' A_{ik}^* + \sum_{k=1}^{K} \sum_{i=1}^{p} G_{ik}^{*\prime} H_{t-i} G_{ik}^*, \tag{2.2}$$

where C_0^*, A_{ik}^*, and G_{ik}^* are $n \times n$ parameter matrices with C_0^* triangular; C_{1k}^* are $J \times n$ parameter matrices; and the summation limit K determines the generality of the process. It should be clear that (2.2) will be positive definite under very weak conditions. Furthermore, this representation is sufficiently general that it includes all positive definite *diagonal* representations and nearly all positive definite *vec* representations. It will be shown to be a particularly convenient representation for estimation and for analysis of simultaneous equations systems. Throughout the paper we will refer to this representation as the *BEKK representation.*[1]

To illustrate the *BEKK* model, consider first the simple GARCH(1,1) model, with $K = 1$ and no exogenous influences[2]:

$$H_t = C_0^{*\prime} C_0^* + A_{11}^{*\prime} \epsilon_{t-1} \epsilon_{t-1}' A_{11}^* + G_{11}^{*\prime} H_{t-1} G_{11}^*. \tag{2.3}$$

In the bivariate case, which is illustrated for both the *vec* and *diagonal* representations earlier, the *BEKK* model becomes

$$H_t = C_0^{*\prime} C_0^* + \begin{bmatrix} a_{11}^* & a_{12}^* \\ a_{21}^* & a_{22}^* \end{bmatrix}' \begin{bmatrix} \epsilon_{1,t-1}^2 & \epsilon_{1,t-1}\epsilon_{2,t-1} \\ \epsilon_{2,t-1}\epsilon_{1,t-1} & \epsilon_{2,t-1}^2 \end{bmatrix} \begin{bmatrix} a_{11}^* & a_{12}^* \\ a_{21}^* & a_{22}^* \end{bmatrix} + \begin{bmatrix} g_{11}^* & g_{12}^* \\ g_{21}^* & g_{22}^* \end{bmatrix}' H_{t-1} \begin{bmatrix} g_{11}^* & g_{12}^* \\ g_{21}^* & g_{22}^* \end{bmatrix}$$

or, suppressing the time subscripts and the GARCH terms,

$$h_{11} = c_{11} + a_{11}^{*2}\epsilon_1^2 + 2a_{11}^* a_{21}^* \epsilon_1 \epsilon_2 + a_{21}^{*2}\epsilon_2^2$$

$$h_{12} = c_{12} + a_{11}^* a_{12}^* \epsilon_1^2 + (a_{21}^* a_{12}^* + a_{11}^* a_{22}^*)\epsilon_1 \epsilon_2 + a_{21}^* a_{22}^* \epsilon_2^2$$

$$h_{22} = c_{13} + a_{12}^{*2}\epsilon_1^2 + 2a_{12}^* a_{22}^* \epsilon_1 \epsilon_2 + a_{22}^{*2}\epsilon_2^2.$$

Comparing this model to the *vec* form of the model, we see that this model economizes on parameters by imposing restrictions both across and within equations. In fact, for $n = 2$ we see that this representation uses only eight parameters, compared to the 18 from the *vec* model (excluding constants).

Before formalizing the relationship between the *BEKK* and *vec* models, we first discuss the identification of the parameters in model (2.3). Proposition

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2.1 shows that under simple and straightforward conditions the parameters in this model, i.e., the parameters in the *BEKK* model with $K = 1$, are identified. Defining two representations to be equivalent if every sequence $\{\epsilon_t\}$ generates the same sequence $\{H_t\}$ for both representations, we have the following proposition.

PROPOSITION 2.1. *Suppose that the diagonal elements in C_0^* are restricted to be positive and that a_{11}^* and g_{11}^* are also restricted to be positive. Then, if $K = 1$, there exists no other C_0^*, A_1^*, or G_1^* in model* (2.3) *that will give an equivalent representation.*

Proof. All proofs are given in the Appendix. ■

In practice, nonnegativity restrictions on parameters are easy to impose, for example, by estimating the square root of the restricted parameter, making identification of the parameters in (2.3) relatively easy for estimation. Also, it should be clear from the proof of Proposition 2.1 that the purpose of the restrictions is to eliminate all other observationally equivalent structures and that there are several other sets of sufficient conditions that could be used in place of those given. For example, as relates to the term $A_1^{*\prime}\epsilon_{t-1}\epsilon_{t-1}^{*}A_1^{*}$, the only other observationally equivalent structure is obtained by replacing A_1^* with $-A_1^*$. The restriction that a_{11}^* be positive could be replaced with the condition that a_{ij}^* be positive for a given i and j, as this condition is also sufficient to eliminate $-A_1^*$ from the set of admissible structures. A final comment is that we decompose the constant matrix into $C_0^{*\prime}C_0^*$ only to ensure positive definiteness; in practice, the elements of C_0^* are not of interest. We chose this decomposition because of its simplicity, but any other identifiable factorization of the constant matrix could be used.

The preceding discussion and illustrations all deal with the *BEKK* model with $K = 1$. Clearly, however, setting $K = 1$ involves imposing restrictions on the model that might not be desirable in practice. The full generality of the *BEKK* representation can be recovered by simply adding more positive semidefinite terms to the variance equation, i.e., by letting $K > 1$ in (2.2), giving in the GARCH(1,1) case

$$H_t = C_0^{*\prime}C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime}\epsilon_{t-1}\epsilon_{t-1}^{\prime}A_{1k}^{*} + \sum_{k=1}^{K} G_{1k}^{*\prime}H_{t-1}G_{1k}^{*}. \tag{2.4}$$

The question of how large K must be for the *BEKK* representation to be "fully general"—i.e., to be equivalent to as many *vec* representations as possible—is answered in part by Proposition 2.2, which gives conditions on the A_{1k}^* and G_{1k}^* matrices that must be satisfied in order to eliminate all the unnecessary restrictions.

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PROPOSITION 2.2. *In order for the GARCH*(1,1) *BEKK model to achieve full generality, the following two necessary conditions must hold*:

(a) *Define* $s = (n(n+1))/2$. *Then, K must be large enough that there are a total of at least* s^2 *distinct parameters in the* A_{1k}^* *matrices.*

(b) *Define* $a_{ij,k}^*$ *to be the* ijth *element of* A_{1k}^*. *Then, there must exist an* A_{1k}^* *matrix that contains either the pair of nonzero elements* $(a_{il,k}^*, a_{jm,k}^*)$ *or the pair of nonzero elements* $(a_{jl,k}^*, a_{im,k}^*)$, *for all* i,j,l,m *between* 1 *and n.*

Similar restrictions hold for the G_{1k}^* *matrices.*

The first condition simply says that if there are fewer parameters in the *BEKK* model than in the *vec* model, then the *BEKK* model is implicitly imposing some unnecessary restrictions. The second condition says that certain pairs of parameters must appear together in an A_{1k}^* matrix for some k in order not to impose extra implicit restrictions. To illustrate, consider the case for $n = 2$. The following set of A_{1k}^* matrices satisfies the conditions in Proposition 2.2:

$$A_{11}^* = \begin{bmatrix} a_{11,1} & a_{12,1} \\ 0 & a_{22,1} \end{bmatrix} \quad A_{12}^* = \begin{bmatrix} a_{11,2} & a_{12,2} \\ a_{21,2} & 0 \end{bmatrix} \quad A_{13}^* = \begin{bmatrix} a_{11,3} & 0 \\ a_{21,3} & a_{22,3} \end{bmatrix}.$$

But the following matrices violate condition (b) of Proposition 2.2 and therefore cannot give a fully general parameterization:

$$A_{11}^* = \begin{bmatrix} a_{11,1} & a_{12,1} \\ 0 & a_{22,1} \end{bmatrix} \quad A_{12}^* = \begin{bmatrix} a_{11,2} & a_{12,2} \\ 0 & 0 \end{bmatrix} \quad A_{13}^* = \begin{bmatrix} a_{11,3} & 0 \\ a_{21,3} & 0 \end{bmatrix}$$

$$A_{14}^* = \begin{bmatrix} 0 & a_{12,4} \\ 0 & a_{22,4} \end{bmatrix}$$

because the pair a_{21} and a_{22} never appears together in any of the A_{1k}^* matrices. This restriction translates into the restriction that the term $\epsilon_{2,t-1}^2$ does not appear in the covariance equation.

Of course, Proposition 2.2 gives only necessary, and not sufficient, conditions for the full generality of the *BEKK* model. Many different sets of sufficient conditions are possible. For example, to look at an extreme case, one set of sufficient conditions is that $K = s$ and none of the A_{1k}^* matrices have any restrictions on its elements. However, this results in identification problems because there are now several equivalent models in the *BEKK* framework. For example, interchanging A_{11}^* and A_{12}^* will give an observationally equivalent structure. In general, an identification problem like this arises in the *BEKK* model whenever $K > 1$, and therefore restrictions must be imposed on the A_{1k}^* and G_{1k}^* matrices to eliminate other equivalent representations. Many different sets of restrictions could be used, but Proposition 2.3 gives a particularly convenient one because the model presented therein is also fully general.[3]

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PROPOSITION 2.3. *Suppose the diagonal elements of C_0^* are restricted to be positive. Consider the class of BEKK models in which $A_{1k_r}^*$, where $k_r = n(r-1)+1,\ldots,nr$, and $r = 1,\ldots,n$, is the matrix obtained by setting the first $r-1$ columns and the first $k_r - n(r-1) - 1$ rows to zero. Suppose also that $a_{nn,k_r}^* > 0\ \forall k_r$ and that similar restrictions are placed on the $G_{1k_r}^*$ matrices. Then, a fully general BEKK model is obtained which has no other equivalent representations in this class.*

Notice that in this representation $K = n^2$. To illustrate, if $n = 2$, then the following set of A_{1k}^* matrices will give a fully general *BEKK* model with no equivalent representations:

$$A_{11}^* = \begin{bmatrix} a_{11,1}^* & a_{12,1}^* \\ a_{21,1}^* & a_{22,1}^* \end{bmatrix} \quad A_{12}^* = \begin{bmatrix} 0 & 0 \\ a_{21,2}^* & a_{22,2}^* \end{bmatrix} \quad A_{13}^* = \begin{bmatrix} 0 & a_{12,3}^* \\ 0 & a_{22,3}^* \end{bmatrix}$$

$$A_{14}^* = \begin{bmatrix} 0 & 0 \\ 0 & a_{22,4}^* \end{bmatrix}.$$

One obvious corollary to this proposition is that for any *BEKK* model with $K = 2$, a sufficient condition to identify the model is that for some i the ith row of A_{12}^* contains only zeros, while the ith row of A_{11}^* contains non-zero elements. Of course, a similar restriction would have to be imposed on G_{12}^*, and the positivity restrictions in Proposition 2.3 would also have to be imposed.

We now turn to a formalization of the relationship between the *BEKK* and *vec* parameterizations. The mathematical relationship between the parameters of these two models can be found by vectorizing both sides of equation (2.4), recognizing that $\mathrm{vec}(ABC) = (C' \otimes A)\mathrm{vec}(B)$:

$$h_t = (C_0^* \otimes C_0^*)'\mathrm{vec}(I_n) + \sum_{k=1}^{K} (A_{1k}^* \otimes A_{1k}^*)'\mathrm{vec}(\epsilon_{t-1}\epsilon_{t-1}') + \sum_{k=1}^{K} (G_{1k}^* \otimes G_{1k}^*)'\mathrm{vec}(H_{t-1}).$$

Therefore,

$$A_1 = \sum_{k=1}^{K} (A_{1k}^* \otimes A_{1k}^*)' \quad \text{and} \quad G_1 = \sum_{k=1}^{K} (G_{1k}^* \otimes G_{1k}^*)',$$

which leads to the following proposition regarding the equivalence of the two models.

PROPOSITION 2.4. *The vec and BEKK parameterizations are equivalent if and only if there exist C_0^*, A_{ik}^*, and G_{ik}^* such that*

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$$C_0 = (C_0^* \otimes C_0^*)' \operatorname{vec}(I_n)$$

$$A_i = \sum_{k=1}^{K} (A_{ik}^* \otimes A_{ik}^*)' \tag{2.5}$$

$$G_i = \sum_{k=1}^{K} (G_{ik}^* \otimes G_{ik}^*)'.$$

One implication of Proposition 2.4 is that the *vec* model implied by any given *BEKK* model is unique, while the converse is not true. The transformation from a *vec* model to a *BEKK* model (when it exists) is not unique, because for a given A_1 the choice of A_{1k}^* is not unique. This can be seen by recognizing that $(A_{1k}^* \otimes A_{1k}^*) = (-A_{1k}^* \otimes -A_{1k}^*)$, so while $A_1 = \sum_{k=1}^{K} (A_{ik}^* \otimes A_{ik}^*)'$ is unique, the choice of A_{ik}^* is not unique.

A second implication of Proposition 2.4 concerns the relationship between the *BEKK* model and the *diagonal* model. In particular, relations (2.5) make it clear that a *diagonal* model is returned from the *BEKK* parameterization if and only if each of the A_{ik}^* and G_{ik}^* matrices are diagonal. It will be shown later that further restrictions can be placed on the diagonal elements of A_{ik}^* and G_{ik}^* in order to obtain a *diagonal* model, but we will postpone that discussion until Proposition 2.6.

A third implication of Proposition 2.4 is the characterization of which *vec* models have *BEKK* representations and which do not. More specifically, we see that the *vec* models excluded from the fully general *BEKK* parameterization are those for which no C_0^*, A_{ik}^*, and G_{ik}^* exist that satisfy relations (2.5). Proposition 2.5 demonstrates that this includes all nonpositive definite *vec* parameterizations. In fact, this is arguably the key feature of the *BEKK* parameterization: positive definite covariance matrices are generated by essentially unrestricted parameterizations. More precisely, we have the following proposition.

PROPOSITION 2.5. *If $H_0, H_{-1}, \ldots, H_{-p+1}$ are all positive definite, then the parameterization of the GARCH equations given in* (2.2) *yields a positive definite H_t for all possible values of ϵ_t if the null space of C_0^* and the null spaces of G_{ik}^*, $i = 1, \ldots, p$, and $k = 1, \ldots, K$, all intersect only at the origin.*

A sufficient condition for this null space criterion to hold is that at least one of the C_0^* or G_{ik}^* be of full rank.

It can also be shown that the *BEKK* model eliminates few, if any, of the interesting positive definite models permitted by the *vec* model. In particular, all positive definite *diagonal vec* models can be written in the *BEKK* framework, so that if one restricts the focus to diagonal models, the *BEKK* model is equally as general as the *vec* model.

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PROPOSITION 2.6. *In the vec model, suppose that the constant part of the covariance matrix is positive definite, so that* $C_0 = \text{vec}(\Omega)$, *where* Ω *is positive definite. Suppose also that* A_i *and* G_i *are diagonal. Then, if* H_t *is positive definite for all possible realizations of* ϵ_t, *there exists a triangular matrix* C_0^* *and diagonal matrices* A_{ik}^* *and* G_{ik}^*, $k = 1, \ldots, n$, *such that*

$$C_0 = \text{vec}(C_0^{*\prime} C_0^*)$$

$$A_i = \sum_{k=1}^{n} (A_{ik}^* \otimes A_{ik}^*)'$$

$$G_i = \sum_{k=1}^{n} (G_{ik}^* \otimes G_{ik}^*)'.$$

This proposition says that the *BEKK* model includes as special cases all possible positive definite linear *diagonal* models and is in this sense "general." The proof of Proposition 2.6 makes it clear that further restrictions can be placed on the A_{ik}^* and G_{ik}^* matrices, beyond just diagonality, without affecting the generality of the *diagonal BEKK* model. In particular, we can restrict each of the A_{ik}^* and G_{ik}^* matrices to be diagonal with the first $n - k$ elements on the diagonals set to zero. By Proposition 2.3, this restricted model will have no other equivalent representations in its class, meaning that we have a fully general linear diagonal model that is both identified and positive definite, making estimation relatively simple.

2.3. Covariance Stationarity

Finally, we turn to a discussion of the necessary and sufficient conditions for covariance stationarity of the multivariate GARCH process. Define the lag operator L such that $L^k w_t \equiv w_{t-k}$. Also, assume that ϵ_t is a doubly infinite sequence and define the multivariate GARCH process as

$$h_t = \sum_{i=1}^{\infty} G(L)^{i-1}[C_0 + A(L)\eta_t]. \tag{2.6}$$

Notice that parameterization (2.6) nests both the *vec* model and the *BEKK* model. To see this, notice that (2.6) implies

$$\begin{aligned} h_t &= C_0 + A(L)\eta_t + \sum_{i=2}^{\infty} G(L)^{i-1}[C_0 + A(L)\eta_t] \\ &= C_0 + A(L)\eta_t + G(L)\sum_{i=1}^{\infty} G(L)^{i-1}[C_0 + A(L)\eta_t] \\ &= C_0 + A(L)\eta_t + G(L)h_t. \end{aligned}$$

Defining $A(L) = A_1 L + A_2 L^2 + \cdots + A_q L^q$ and $G(L) = G_1 L + G_2 L^2 + \cdots + G_p L^p$ gives *vec* model (2.1), and defining $A(L) = \sum_{k=1}^{K} (A_{1k}^* \otimes$

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$A_{1k}^{*})'L + \cdots + \sum_{k=1}^{K} (A_{qk}^{*} \otimes A_{qk}^{*})'L^{q}$ and $G(L) = \sum_{k=1}^{K} (G_{1k}^{*} \otimes G_{1k}^{*})'L + \cdots + \sum_{k=1}^{K} (G_{pk}^{*} \otimes G_{pk}^{*})'L^{p}$ gives *BEKK* parameterization (2.2).

PROPOSITION 2.7. *Suppose the process* $\{\epsilon_t\}$ *is a doubly infinite sequence and equation* (2.6) *defines the GARCH process. Then,* $\{\epsilon_t\}$ *is covariance stationary if and only if all the eigenvalues of* $A(1) + G(1)$ *are less than one in modulus.*

This implies that in the *vec* model $\{\epsilon_t\}$ is covariance stationary if and only if the eigenvalues of $\sum_{i=1}^{q} A_i + \sum_{i=1}^{p} G_i$ are less than one in modulus. Also, in the *BEKK* model, $\{\epsilon_t\}$ is covariance stationary if and only if all the eigenvalues of $\sum_{i=1}^{q} \sum_{k=1}^{K} (A_{ik}^{*} \otimes A_{ik}^{*}) + \sum_{i=1}^{p} \sum_{k=1}^{K} (G_{ik}^{*} \otimes G_{ik}^{*})$ are less than one in modulus. It is evident from the proof to Proposition 2.7 that the unconditional covariance matrix, when it exists, is given by

$$E(\eta_t) = [I - A(1) - G(1)]^{-1} \operatorname{vec} C_0. \tag{2.7}$$

Focusing on the GARCH(1,1) model, this implies that the unconditional covariance matrix in the *vec* model is

$$E(\eta_t) = [I - A_1 - G_1] C_0 \tag{2.7a}$$

and in the *BEKK* model with $K = 1$ it is

$$E(\eta_t) = [I - (A_{11}^{*} \otimes A_{11}^{*})' - (G_{11}^{*} \otimes G_{11}^{*})']^{-1} \operatorname{vec} C_0^{*\prime} C_0^{*}. \tag{2.7b}$$

Several other implications are also apparent. For example, the *diagonal vec* model is stationary if and only if the sums $a_{ii} + g_{ii}$ are less than one for all i, and the *diagonal BEKK* model is stationary if and only if $\sum_{k=1}^{n} (a_{ii,k}^{*2} + g_{ii,k}^{*2}) < 1$ for all i. However, it is important to recognize that it is only in the case of *diagonal* models that the stationarity properties are determined solely by the diagonal elements of the A_{ik}^{*} and G_{ik}^{*} (or A_i and G_i) matrices. In non-*diagonal BEKK* models, for example, it is possible to have diagonal elements exceeding one yet the process be stationary. For example, in the GARCH(0,1) model with $K = 1$ and

$$A_{11}^{*} = \begin{bmatrix} 1.1 & 0.2 \\ -0.2 & 0.7 \end{bmatrix},$$

all four eigenvalues of $(A_{11}^{*} \otimes A_{11}^{*})$ are 0.81. By Proposition 2.7, this process is stationary even though the diagonal elements of A_{11}^{*} are not both less than one.

3. THE REGRESSION MODEL

Applying *BEKK* representation (2.2) to a simultaneous regression model with the second moments appearing in the structural mean equations gives the multivariate GARCH-in-mean model[4]

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$$\epsilon_t = \Gamma y_t + Bx_t + \Lambda \tilde{h}_t$$

$$\epsilon_t | \Im_{t-1} \sim N(0, H_t) \tag{3.1}$$

$$H_t = C_0^{*\prime} C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime} \epsilon_{t-1} \epsilon_{t-1}' A_{1k}^* + \sum_{k=1}^{K} G_{1k}^{*\prime} H_{t-1} G_{1k}^*,$$

where $\Gamma_{n\times n}$, $B_{n\times J}$, and $\Lambda_{n\times s}$ are parameter matrices; y_t $(n \times 1)$ are endogenous variables; x_t $(J \times 1)$ are weakly exogenous and lagged dependent variables; the second moments in the mean equation, $\tilde{h}_t = vech(H_t)$ and $vech(H)$ stacks the lower triangle of matrix H, are predetermined but not weakly exogenous variables.[5] These moments, $\tilde{h}_t$, are not weakly exogenous for Γ, B, and Λ because the information matrix is not block-diagonal between these parameters and the parameters in the marginal distribution of $\tilde{h}_t$. This can be seen by examining the simplest of cases—the GARCH(0,0) model with no intercepts in the mean equations. If $H_t = C_0^{*\prime} C_0^* \equiv \Delta$, then

$$\mathrm{I}_{\Delta\lambda} = (\mathrm{vec}\,\Delta \otimes \Gamma'^{-1})\Delta^{-1}\Gamma^{-1}\Lambda(I \otimes \iota) \neq 0,$$

where $\mathrm{I}_{\Delta\lambda}$ is the block of the information matrix that corresponds to the interactions between $\mathrm{vec}(\Delta)$ and λ, ι is a $n^2 \times 1$ vector of ones, and $\lambda \equiv \mathrm{vec}(\Lambda)$. Alternatively, notice that in the general GARCH(1,1) model $\tilde{h}_t$ is a function of ϵ_{t-1}, and ϵ_{t-1} are functions of the same parameters as $\tilde{h}_t$. So the parameters in the mean equations cannot be estimated without estimating the variance equations. Hence, there is no sequential cut (Engle et al., 1983).

The reduced form of (3.1) will also have a multivariate GARCH representation because nonsingular linear combinations of multivariate GARCH models are GARCH.

PROPOSITION 3.1. *If ϵ_t is a multivariate GARCH process and P is a nonsingular matrix, then $P\epsilon_t$ is also a multivariate GARCH process of the same order.*

One implication of this proposition is that if GARCH is placed on the structural errors as in (3.1), then letting $P = \Gamma^{-1}$, we see that the reduced form will also have GARCH errors with the same orders. In particular, we have

$$\begin{aligned} y_t &= -\Gamma^{-1}Bx_t - \Gamma^{-1}\Lambda\tilde{h}_t + \Gamma^{-1}\epsilon_t \\ &= \Pi_1 x_t + \Pi_2 \tilde{h}_t + \nu_t \end{aligned} \tag{3.2}$$

$$\nu_t | \Im_{t-1} \sim N(0, \Gamma^{-1} H_t \Gamma^{-1\prime}).$$

Furthermore, if the conditions for positive definiteness were satisfied in the structure, then they will be satisfied in the reduced form. Several other implications of this proposition become readily apparent from the following simple relations between the structural and reduced form parameters, which are

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developed in the proof to Proposition 3.1: For the *BEKK* parameterization, the relationships are

$$C^*_{rf,0}\Gamma' = C^*_{s,0}$$

$$\Gamma A^*_{rf,ik} = A^*_{s,ik}\Gamma$$

$$\Gamma G^*_{rf,ik} = G^*_{s,ik}\Gamma,$$

and for the *vec* parameterization they are

$$(\Gamma \otimes \Gamma)C_{rf,0} = C_{s,0}$$

$$(\Gamma \otimes \Gamma)A_{rf,i} = A_{s,i}(\Gamma \otimes \Gamma)$$

$$(\Gamma \otimes \Gamma)G_{rf,i} = G_{s,i}(\Gamma \otimes \Gamma),$$

where the *rf* subscript refers to reduced form parameters and the *s* subscript refers to structural parameters. For example, one implication is that if the GARCH error process is placed on the reduced form model, then the structural model must have GARCH errors. Also, if any part of the structural (reduced form) covariance matrix follows a GARCH process, then in general the complete reduced form (structural) covariance matrix will follow a GARCH process.

Identification of the structural coefficients in (3.1) using $\tilde{h}_t$ cannot proceed as in the standard simultaneous equations model both because linear combinations of the structural equations in the system will change the definition of $\tilde{h}_t$ and because $\tilde{h}_t$ is unobservable and consistent estimates of $\tilde{h}_t$ can only be obtained if the model is identified. But rewriting $\Lambda\tilde{h}_t$ in terms of the reduced form covariance matrix permits standard identification procedures.[6] To do this, we first note that because the parameters in the conditional covariance matrices are identified (see Section 2.2), we can get consistent estimates of the reduced form covariances $h^*_t = vech(\Gamma^{-1}H_{t-1}\Gamma^{-1\prime})$. Using

$$\tilde{h}_t = vech(H_t) = R\,\mathrm{vec}(H_t) = R(\Gamma \otimes \Gamma)\mathrm{vec}(\Gamma^{-1}H_t\Gamma^{-1\prime})$$

$$= R(\Gamma \otimes \Gamma)Sh^*_t,$$

where $R_{s\times n^2}$ is the selection matrix such that $vech(P) = R\,\mathrm{vec}(P)$, and $S_{n^2\times s}$ is the expansion matrix such that $\mathrm{vec}(P) = S\,vech(P)$ for any square matrix $P_{n\times n}$, gives

$$\Lambda\tilde{h}_t = \Lambda R(\Gamma \otimes \Gamma)Sh^*_t = \Lambda^* h^*_t. \tag{3.3}$$

This gives the new system

$$\Gamma y_t + Bx_t + \Lambda^* h^*_t = \epsilon_t, \tag{3.4}$$

where h^*_t, though unobservable, is consistently estimable and is invariant with respect to linear combinations of the structural equations. Identifica-

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tion of the new system of (3.4) now reduces to the standard problem of identification of a linear system (with variables y_t, x_t, h_t^*) that is nonlinear in the parameters. The nonlinearities arise because $\Lambda^* = \Lambda R(\Gamma \otimes \Gamma)S$ is a nonlinear function of Λ and Γ. So necessary and sufficient conditions for identification in this model must be given by conditions involving (Γ, B, Λ^*), not (Γ, B, Λ), as might at first seem natural. In other words, restrictions on Λ must first be transformed into restrictions involving Λ^* before they can be used in identification.

To illustrate the problems that can arise if $\Lambda\tilde{h}_t$ is used in identification instead of $\Lambda^* h^*$, consider the following bivariate system of equations (ignoring time subscripts):

$$\begin{aligned} y_1 &= \beta_1 y_2 + \lambda_{11} h_{11} + \lambda_{13} h_{22} + \epsilon_1 \\ y_2 &= \lambda_{21} h_{11} + \lambda_{22} h_{12} + \lambda_{23} h_{22} + \epsilon_2. \end{aligned} \tag{3.5}$$

Identification of this system means that premultiplying by any nondiagonal matrix does not give an observationally equivalent structure. However, premultiplying this system by the matrix

$$\Upsilon = \begin{bmatrix} 1 & v \\ 0 & 1 \end{bmatrix}$$

gives the new system

$$\begin{aligned} y_1 &= (\beta_1 - v)y_2 + (\lambda_{11} + v\lambda_{21})h_{11} + (\lambda_{13} + v\lambda_{23})h_{22} \\ &\quad + v\lambda_{22}h_{12} + \epsilon_1 + v\epsilon_2 \\ y_2 &= \lambda_{21} h_{11} + \lambda_{22} h_{12} + \lambda_{23} h_{22} + \epsilon_2. \end{aligned} \tag{3.6}$$

When estimating system (3.6), the GARCH-in-mean terms will be measuring the conditional variances and covariances of the residuals in transformed system (3.6), not those in original system (3.5). The residual in the first transformed equation is $\epsilon_1 + v\epsilon_2$, which has conditional variance $h_{11}^\diamond = h_{11} + 2vh_{12} + v^2 h_{22}$, and the residual in the second transformed equation is still ϵ_2, which has conditional variance $h_{22}^\diamond = h_{22}$. The conditional covariance between the transformed residuals is $h_{12}^\diamond = h_{12} + vh_{22}$. Solving these three relationships for h_{11}, h_{12}, and h_{22} and then substituting into (3.6) gives

$$\begin{aligned} y_1 &= (\beta_1 - v)y_2 + (\lambda_{11} + v\lambda_{21})h_{11}^\diamond + v(\lambda_{22} - 2\lambda_{11} - 2v\lambda_{21})h_{12}^\diamond \\ &\quad + [\lambda_{13} + v\lambda_{23} + v^2(\lambda_{11} - \lambda_{22}) + v^3\lambda_{21}]h_{22}^\diamond + \epsilon_1 + v\epsilon_2 \\ y_2 &= \lambda_{21} h_{11}^\diamond + (\lambda_{22} - 2v\lambda_{21})h_{12}^\diamond + (\lambda_{23} - v\lambda_{22} + v^2\lambda_{21})h_{22}^\diamond + \epsilon_2. \end{aligned}$$

Setting

$$v = \frac{\lambda_{22} - 2\lambda_{11}}{2\lambda_{21}}$$

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gives an observationally equivalent structure to (3.5), indicating that the first equation in (3.5) is not identified. However, using traditional rank and order conditions on (3.5) would erroneously lead us to conclude that system (3.5) is identified. This happens because the definition of $\tilde{h}_t$ changes in different structures, so $\tilde{h}_t$ is no help in ruling out equivalent structures. But reduced form variances and covariances are invariant to different structures, so a priori restrictions on the reduced form variances and covariances in the mean equations can be used for identification purposes. Using (3.3) to rewrite (3.5) in terms of reduced form variances and covariances gives the new system

$$\begin{aligned} y_1 &= \beta_1 y_2 + \lambda_{11} h_{11}^* - 2\beta_1 \lambda_{11} h_{12}^* + (\lambda_{13} + \beta_1^2 \lambda_{11}) h_{22}^* + \epsilon_1 \\ y_2 &= \lambda_{21} h_{11}^* + (\lambda_{22} - 2\beta_1 \lambda_{21}) h_{12}^* + (\lambda_{23} - \beta_1 \lambda_{22} + \beta_1^2 \lambda_{21}) h_{22}^* + \epsilon_2, \end{aligned} \tag{3.7}$$

where h_{ij}^* are the reduced form variances and covariances. In this system, it is straightforward to show that the same linear combination of equations as earlier results in an observationally equivalent structure to (3.7), meaning system (3.7) is not identified. However, it is clear that restrictions on the reduced form variances and covariances in system (3.7) can identify the system. For example, if it is known that the coefficient on h_{12}^* is zero in (3.7), then the system is identified. To conclude, rank and order conditions can be used with h_t^* treated as predetermined, giving standard identification results. This is often natural because h_t^* is the set of forecast variances and covariances of the endogenous variables. The use of standard rank and order conditions on h_t may give incorrect identification conclusions.[7]

4. SYSTEM ESTIMATION

On the one hand, if Λ equals zero in model (3.1) and the system is identified, the equations can be consistently estimated with two-stage or three-stage least squares, ignoring the GARCH error structure, because the reduced form estimates of y_t remain uncorrelated with ϵ_t. However, more efficient estimates can be obtained by accounting for the error structure, for example, by using full information maximum likelihood or an instrumental variables estimator. On the other hand, if $\Lambda \neq 0$, then full information methods are generally required to obtain efficient or often even consistent estimates of the models' parameters. Before discussing these methods further, we will first present the likelihood function.

Although the error vectors ϵ_t are conditionally multivariate-normally distributed, their outer products $\epsilon_t \epsilon_t'$ are not independent through time. For example, $\epsilon_{1,t}^2$ need not be uncorrelated with $\epsilon_{1,t-1}^2$. Hence, the joint distribution of $(\epsilon_1, \epsilon_2, \ldots, \epsilon_T)$, where T is the number of observations, need not be multivariate-normally distributed. But the joint density is the product of all the conditional densities, so the log likelihood function of the joint distribution is the sum of all the log likelihood functions of the conditional distri-

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butions, i.e., the sum of the logs of the multivariate-normal distribution. So if we are estimating reduced form model (3.2), then letting L_t be the log likelihood of observation t, and L be the joint log likelihood, gives

$$L = \sum_{t=1}^{T} L_t$$

$$L_t = \frac{n}{2}\ln(2\pi) - \frac{1}{2}\ln|H_{rf,t}| - \frac{1}{2}\nu_t' H_{rf,t}^{-1}\nu_t. \tag{4.1}$$

Proposition 3.1 says that the structural model will also follow a multivariate GARCH process, so applying the transformation $\epsilon_t = \Gamma\nu_t$ gives

$$L_t = \frac{n}{2}\ln(2\pi) + \ln|\Gamma| - \frac{1}{2}\ln|H_t| - \frac{1}{2}\epsilon_t' H_t^{-1}\epsilon_t. \tag{4.2}$$

Of course, these likelihood functions are not fully specified until we make assumptions on initial conditions. A reasonable set of assumptions is that all presample data be fixed at their unconditional expectation. So, for example, $\epsilon_0\epsilon_0'$ is assumed to equal its unconditional expectation, given in equation (2.7).

Note that because no reference is made in (4.1) or (4.2) to the functional form chosen for the conditional covariance matrix, the results of this section apply whether the *vec* or *BEKK* parameterization is chosen. In either case, however, the models are large and complex, leading one to question how flat the likelihood function is with respect to many of the parameters in the covariance equations. A reasonable procedure is to estimate a restricted model such as the diagonal model or the *BEKK* model with $K = 1$ and then use a Lagrange multiplier test to examine the validity of the restriction.

Define Ξ to be the nonredundant parameters in the covariance equations, so that

$$\Xi' = [(RC_0)', (RA_1S)', \ldots, (RA_qS)', (RG_1S)', \ldots, (RG_pS)']$$

for the *vec* representation or

$$\Xi' = [C_0^{*\prime}, A_{11}^{*\prime}, \ldots, A_{qK}^{*\prime}, G_{11}^{*\prime}, \ldots, G_{pK}^{*\prime}]$$

for the *BEKK* representation. Then, the maximum likelihood estimator for the parameters in the structural model, Γ, B, Λ, and Ξ, is found by maximizing (4.2) with respect to these parameters. Unfortunately, the properties of maximum likelihood estimators in GARCH models are still open to debate, so little can confidently be said about the asymptotic properties of this estimator. Furthermore, the properties of maximum likelihood estimates of GARCH-in-mean models ($\Lambda \neq 0$) have not even been addressed in the literature. Instead of attempting to close the book on these issues here, we refer the reader to the papers by Bollerslev and Wooldridge (1992), Lee and Hansen (1994), Lumsdaine (1991), Pagan and Sabau (1987), or Weiss (1986),

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which all address (with varying degrees of success) the properties of maximum likelihood and quasi-maximum likelihood estimates in univariate ARCH models.

Letting $\theta' = [(\text{vec}\,\Gamma)', (\text{vec}\,B)', (\text{vec}\,\Lambda)', (\text{vec}\,\Xi)']$ and differentiating (4.2) with respect to θ gives

$$\frac{\partial L_t}{\partial \theta} = \frac{1}{2}\left(\frac{\partial h_t}{\partial \theta}\right)'(H_t^{-1} \otimes H_t^{-1})\text{vec}(\epsilon_t\epsilon_t' - H_t) - \left(\frac{\partial\,\text{vec}\,\Gamma}{\partial \theta}\right)'\text{vec}(\Gamma^{-1\prime}) - \left(\frac{\partial \epsilon_t}{\partial \theta}\right)'H_t^{-1}\epsilon_t.$$

Noting, for example, that[8]

$$\frac{\partial h_t}{\partial\,\text{vec}\,B} = \sum_{i=1}^{q} A_i[(I \otimes \epsilon_{t-i}) + (\epsilon_{t-i} \otimes I)]\frac{\partial \epsilon_{t-i}}{\partial\,\text{vec}\,B} + \sum_{i=1}^{p} G_i \frac{\partial h_{t-i}}{\partial\,\text{vec}\,B}$$

and that

$$\frac{\partial \epsilon_t}{\partial\,\text{vec}\,B} = (x_t' \otimes I) + \Lambda\frac{\partial h_t}{\partial\,\text{vec}\,B}$$

we see that calculation of $(\partial h_t)/(\partial\theta)$ is complicated by the fact that $(\partial h_t)/(\partial\theta)$ depends on $(\partial h_{t-1})/(\partial\theta)$ if $G_i \neq 0$ for any i or if $\Lambda \neq 0$. Hence, the use of analytical derivatives would require calculating $(\partial h_t)/(\partial\theta)$ recursively, with $(\partial h_0)/(\partial\theta)$ assumed to be independent of θ. Because of this problem and because the derivatives $(\partial h_t)/(\partial\theta)$ and $(\partial\epsilon_t)/(\partial\theta)$ are so cumbersome, nonlinear maximization methods and numerical derivatives seem appropriate. Many nonlinear maximization methods are available, but a particularly convenient one and one that the authors have found useful in practice is the Berndt, Hall, Hall, and Hausman (1974) algorithm, which is an iterative method of calculating the optimal parameters in which the updating term is found by a regression of a vector of 1's on the scores $(\partial L)/(\partial\theta)$:

$$\theta^{i+1} = \theta^i + \tau_i(S'S)^{-1}S'\iota.$$

Here, $[S]_{tk} = (\partial L_t)/\partial\theta_k$, i represents the iteration number, ι is the vector of 1's, and τ_i is the step length that is calculated at each iteration by a line search. There are three features of the BHHH algorithm that make it particularly advantageous in these models. First, it is easy to use in practice, because its use requires little more than a subroutine to compute numerical derivatives and a subroutine to compute OLS regression parameters. Second, under normality, the $(S'S)^{-1}$ from the final iteration can be used as a consistent estimate of the variance–covariance matrix of the parameters. And third, Lagrange multiplier statistics are easily computed as T times the R^2 of the regression in the first step of the BHHH iteration, starting at the estimates under the null. This provides a particularly easy way to test the valid-

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ity of any restrictions that might have been imposed during estimation, such as diagonality, $p = q = 1$, or $\Lambda = 0$, and suggests the following model building strategy: Begin by estimating a *diagonal BEKK* model with $K = 1$, then use the Lagrange multiplier test described above to examine whether the diagonality restriction is valid. If not, then either additional factors can be added or additional terms added to the first factor. Alternatively, if the *vech* model is chosen, then one could begin by estimating a diagonal *vech* model and then use the preceding LM test to examine the validity of the restriction. If rejected, then the appropriate nondiagonal terms should be added. This "bottom-up" model building procedure is easy to use in practice because *diagonal* models are not difficult to estimate, and once they are estimated the LM tests are very easy to compute using the procedure described above. Furthermore, if the restrictions are rejected, then the diagonal model provides an obvious set of starting values to use in the nonlinear estimation of the unrestricted model. However, as usual, this model building procedure is only guaranteed to be consistent in certain special cases.

Of course, BHHH is only one of several possible optimization algorithms, each of which has its advantages. For example, it is widely recognized that as one gets closer to the optimum the benefits to calculating the expected value of the Hessian become enormous, relative to using the outer product of the scores (as in BHHH). This is both because the convergence rate tends to increase enough to offset the extra effort required and because the estimated standard errors tend to become more accurate.

Note that in the reduced form model with $\Lambda = 0$ the information matrix is block-diagonal between the parameters in the mean equations and the parameters in the covariance equations (Kraft and Engle, 1983). This means that efficient estimates of Π_1 can be calculated independently of Ξ, given only $\sqrt{T}$-consistent estimates of Ξ. Similarly, efficient estimates of Ξ can be calculated independently of Π_1, given only $\sqrt{T}$-consistent estimates of Π_1. This suggests the following estimation procedure. First, obtain consistent estimates of Π_1 (call then $\tilde{\Pi}_1$) from a seemingly unrelated regression.[9] Then, maximize likelihood (4.1) with respect to vec Ξ, given $\tilde{\Pi}_1$, to get an estimate of vec Ξ, say vec $\hat{\Xi}$. Finally, maximize (4.1) again, this time with respect to Π_1 (given vec $\hat{\Xi}$), imposing the structural restrictions, to get $\hat{\Pi}_1$. ($\hat{\Pi}_1$,vec $\hat{\Xi}$) are asymptotically equivalent to the maximum likelihood estimates. If this procedure is iterated and converges, the estimates will solve the first-order conditions for full information maximum likelihood estimation.

5. CONCLUSION

This paper extends Engle's (1982) ARCH model and Bollerslev's (1986) GARCH model to a multivariate setting. To parameterize the multivariate process so that positive definiteness is ensured, a new formulation is presented and compared with that used in much of the existing multivariate

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ARCH literature. Equivalence relations between these parameterizations are derived, conditions for covariance stationarity are presented, and the relationship between the reduced form and structural models is analyzed. Maximum likelihood estimation of the system is then discussed, though we have little to say about the properties of this estimator. This area is perhaps one of the most important areas for future research in multivariate ARCH modeling. Very little is currently known about the properties of maximum likelihood estimators in *univariate* GARCH models, let alone in *multivariate* GARCH-*in-mean* models, despite the fact that this estimator permeates the multivariate GARCH-in-mean literature.

NOTES

1. This acronym comes from an earlier version of this paper, which synthesized the work on multivariate ARCH models by Yoshi Baba, Rob Engle, Dennis Kraft, and Ken Kroner.

2. For the remainder of this section of the paper, the terms representing the exogenous influences will be dropped in order to shorten some of the equations and proofs. From the structure of the proofs, it should be clear that the extension would involve simply repeating some of the steps of the proofs or adding another term to the proofs.

3. To keep the notation as simple as possible, we focus on GARCH(1,1) models. The generalization to higher-order models is trivial.

4. Again, we focus here on the GARCH(1,1) model to keep the notation simple. Also, we focus only on the model with variances and covariances in the mean equation. The implications of using standard deviations or other functions of the variances, as in Kroner and Lastrapes (1993), will be the same.

5. The *vech* operator is the vector-half operator, which stacks the lower triangular portion of a matrix.

6. We would like to acknowledge an anonymous referee for suggesting this solution.

7. Notice that, because the unconditional distribution is not normal, information in the higher moments and in the information matrix can also be used to help identify the model.

8. This derivative assumes the *vec* model was used. A similar expression holds for the *BEKK* model.

9. See equation (2.7) for the formula for the unconditional covariance matrix.

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APPENDIX

Proof for Proposition 2.1. First, we show that if the diagonal elements of C_0^* are restricted to be positive, then the C_0^* matrix is identified: If the diagonal elements of C_0^* are positive, then we know that $C_0^{*\prime} C_0^*$ is a positive definite matrix. But by Proposition 58 and Remark 34 of Dhrymes (1984, pp. 68–69), the decomposition of a positive definite matrix into the product of a triangular matrix and its transpose always exists, and this decomposition is unique if the diagonal terms are restricted to be positive.

Next, we show that $a_{11}^* > 0$ is sufficient to identify the A_{11}^* matrix. Straightforward algebra reveals that the lmth element of H_t is

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$$h_{lm,t} = c + \sum_{i=1}^{n}\sum_{j=1}^{n} a_{il}^{*}a_{jm}^{*}\epsilon_{i,t-1}\epsilon_{j,t-1} + \sum_{i=1}^{n}\sum_{j=1}^{n} g_{il}^{*}g_{jm}^{*}h_{ij,t-1}.$$

So, for example, ignoring the constant and the GARCH terms, the (1,1) element of H_t is

$$h_{11,t} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{i1}^{*}a_{j1}^{*}\epsilon_{i,t-1}\epsilon_{j,t-1}.$$

So the coefficient attached to $\epsilon_{1,t-1}^{2}$ is a_{11}^{*2}, meaning that a_{11}^{*} is identified, up to its sign. Restricting it to be positive identifies this term. Next, the coefficient attached to $\epsilon_{1,t-1}\epsilon_{j,t-1}$ in this equation is $(a_{11}^{*}a_{j1}^{*} + a_{j1}^{*}a_{11}^{*}) = 2a_{11}^{*}a_{j1}^{*}$. So a_{j1}^{*} is identified for all j because a_{11}^{*} is identified. So we have that the first column of A_{11} is identified. Next, the (1,2) element of H_t is, again ignoring the constant and the GARCH terms,

$$h_{12,t} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{i1}^{*}a_{j2}^{*}\epsilon_{i,t-1}\epsilon_{j,t-1}.$$

The coefficient attached to $\epsilon_{1,t-1}\epsilon_{1,t-1}$ in this equation is $(a_{11}^{*}a_{12}^{*} + a_{11}^{*}a_{12}^{*})$. Because a_{11}^{*} is identified, a_{12}^{*} must also be identified. Similar logic can be used to show that all the other terms in A_{11}^{*} are identified, completing the proof that the matrix A_{11} is identified.

The identification of the G_{11} matrix follows an identical argument as above and is therefore not presented here. ■

Proof for Proposition 2.2.

(a) The proof here is trivial. Ignoring the constants and GARCH terms, there are a total of s^2 variables on the right-hand side of the covariance equations, so if we have fewer than s^2 parameters, some restrictions are implicitly being imposed.

(b) Elementary algebra reveals that the lmth element of H_t in the *BEKK* model with $K > 1$ is

$$h_{lm,t} = c + \sum_{k=1}^{K}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{il,k}^{*}a_{jm,k}^{*}\epsilon_{i,t-1}\epsilon_{j,t-1} + \sum_{k=1}^{K}\sum_{i=1}^{n}\sum_{j=1}^{n} g_{il,k}^{*}g_{jm,k}^{*}h_{ij,t-1}.$$

So the coefficient attached to $\epsilon_{i,t-1}\epsilon_{j,t-1}$ in the $h_{lm,t}$ equation is

$$\sum_{k=1}^{K} a_{il,k}^{*}a_{jm,k}^{*} + a_{jl,k}^{*}a_{km,k}^{*}$$

if $i \neq j$ and

$$\sum_{k=1}^{K} a_{il,k}^{*}a_{im,k}^{*}$$

if $i = j$. Clearly, the term $\epsilon_{i,t-1}\epsilon_{j,t-1}$ will drop out of the $h_{lm,t}$ equation if there is no matrix A_{1k}^{*} that contains either both of $a_{il,k}^{*}$ and $a_{jm,k}^{*}$ or both of $a_{jl,k}^{*}$ and $a_{im,k}^{*}$. Hence, for the model to be fully general, we require that either the pair $(a_{il,k}^{*}, a_{jm,k}^{*})$ or the pair $(a_{jl,k}^{*}a_{im,k}^{*})$ appear in a A_{1k} matrix for some k. ■

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Proof for Proposition 2.3. First, in the proof for Proposition 2.1, we show that the conditions on C_0^* are sufficient to ensure that the elements of C_0^* are identified. Also, as in the proof to Proposition 2.1, we focus our proof on the ARCH terms because the proof for the GARCH terms is directly analogous. The proof proceeds in a manner similar to the proof for Proposition 2.1.

From the proof for Proposition 2.2, we have

$$h_{lm,t} = c + \sum_{k_r=1}^{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{il,k_r}^* a_{jm,k_r}^* \epsilon_{i,t-1}\epsilon_{j,t-1} + \sum_{k_r=1}^{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} g_{il,k_r}^* g_{jm,k_r}^* h_{ij,t-1}$$

and that the coefficient on $\epsilon_{1,t-1}^2$ in the $h_{11,t}$ equation is $\sum_{k_r=1}^{n^2} a_{11,k_r}^{*2}$, which is just $a_{11,1}^{*2}$ because the restrictions on the $A_{1k_r}^*$ matrices ensure that $a_{11,k_r}^* = 0$ for all $k_r > 1$. Therefore, $a_{11,1}^*$ is identified up to its sign. Next, the coefficient on $\epsilon_{1,t-1}\epsilon_{j,t-1}$ $(j \neq 1)$ in the $h_{11,t}$ equation is $2\sum_{k_r=1}^{n^2} a_{11,k_r}^* a_{j1,k_r}^*$, which is $2a_{11,1}^* a_{j1,1}^*$ because the restrictions on the $A_{1k_r}^*$ matrices ensure that $a_{11,k_r}^* = 0$ for all $k_r > 1$. Therefore, $a_{j1,1}^*$, $j = 2, \ldots, n$, are identified (up to their sign) because $a_{11,1}^*$ is identified. The coefficient on $\epsilon_{2,t-1}^2$ in the $h_{11,t}$ equation is $\sum_{k_r=1}^{n^2} a_{21,k_r}^{*2}$, which becomes $(a_{21,1}^{*2} + a_{21,2}^{*2})$ because the zero restrictions on the other A_{1k_r} matrices imply that $a_{21,k_r}^* = 0$ for all $k_r > 2$. Therefore, $a_{21,2}^*$ is identified (up to its sign) because $a_{21,1}^2$ was previously identified. Proceeding similarly, the coefficient on $\epsilon_{2,t-1}\epsilon_{j,t-1}$ (if $j \neq 2$) is $2\sum_{k_r=1}^{n^2} a_{21,k_r}^* a_{j1,k_r}^*$, which becomes $2(a_{21,1}^* a_{j1,1}^* + a_{21,2}^* a_{j1,2}^*)$ because the zero restrictions on the other $A_{1k_r}^*$ matrices imply that $a_{21,k_r}^* = 0$ for all $k_r > 2$. So $a_{j1,2}^*$, $j = 3,4, \ldots, n$, are identified (up to sign), because all the other terms in $2(a_{21,1}^* a_{j1,1}^* + a_{21,2}^* a_{j1,2}^*)$ were already identified. Similar logic can be used to identify a_{j1,k_r}^*, $k_r = 3, \ldots, n$. So we have that the first columns of the $A_{1k_r}^*$ matrices, $k_r = 1, \ldots, n$, are identified (except that each column can be multiplied by -1 without changing the model).

To identify the second columns of the $A_{1k_r}^*$ matrices, $k_r = 1, \ldots, n$, we look at the $h_{12,t}$ equation. The coefficient on $\epsilon_{1,t-1}^2$ is $\sum_{k_r=1}^{n^2} a_{11,k_r}^* a_{12,k_r}^*$, which equals $a_{11,1}^* a_{12,1}^*$ because the zero restrictions on the $A_{1k_r}^*$ matrices ensure that $a_{11,k_r}^* = 0$ for all $k_r > 1$. This identifies $a_{12,1}^*$ because $a_{11,1}^*$ is identified. The coefficient on $\epsilon_{1,t-1}\epsilon_{j,t-1}$, after accounting for the zero restrictions, is $a_{11,1}^* a_{j2,1}^* + a_{j1,1}^* a_{12,1}^*$, which identifies $a_{j2,1}^*$, $j = 2, \ldots, n$, because the other terms are identified. So the second column of A_{11}^* is identified. The coefficient on $\epsilon_{2,t-1}^2$ is $a_{21,1}^* a_{22,1}^* + a_{21,2}^* a_{22,2}^*$, which identifies $a_{22,2}^*$, and the coefficient on $\epsilon_{2,t-1}\epsilon_{j,t-1}$, $j = 3, \ldots, n$, is $a_{21,1}^* a_{j2,1}^* + a_{j1,1}^* a_{22,1}^* + a_{21,2}^* a_{j2,2}^* + a_{22,2}^* a_{j1,2}^*$, which identifies $a_{j2,2}^*$, $j = 3, \ldots, n$. Therefore, the second column of A_{12}^* is identified, and continuing along these lines will identify the second columns of the remaining A_{1k_r} matrices, $k_r = 1, \ldots, n$.

Proceeding similarly will identify the remaining columns of the matrices $A_{11}^*, \ldots, A_{1n}^*$ up to sign. The restriction that $a_{nn,k_r}^* > 0$ will eliminate the sign problem, and these matrices are now identified. Identification of the remaining matrices proceeds analogously. For example, the elements of the $h_{22,t}$ equation will be used to identify the second column of the matrices $A_{1,n+1}^*, \ldots, A_{1,2n}^*$, and the elements of the $h_{23,t}$ equation will be used to identify the third column of these matrices.

Finally, notice that every coefficient in every equation has a free parameter. Therefore, each coefficient can take on any value permitted by the *BEKK* model, and adding other $A_{1k_r}^*$ matrices cannot give a model that was previously being precluded. So the model is fully general. ■

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Proof for Proposition 2.4. For simplicity, let $p = q = 1$. Then, the *vec* parameterization becomes

$$h_t = \left[C_0 \vdots A_1 \vdots G_1\right]\begin{bmatrix} 1 \\ \eta_{t-1} \\ h_{t-1} \end{bmatrix} \tag{A.1}$$

and the *BEKK* parameterization becomes

$$H_t = C_0^{*\prime} C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime} \epsilon_{t-1} \epsilon_{t-1}' A_{1k}^* + \sum_{k=1}^{K} G_{1k}^{*\prime} H_{t-1} G_{1k}^*. \tag{A.2}$$

Vectorizing,

$$\begin{aligned} h_t &= \operatorname{vec}(C_0^{*\prime} C_0^*) + \operatorname{vec} \sum_{k=1}^{K} A_{1k}^{*\prime} \epsilon_{t-1} \epsilon_{t-1}' A_{1k}^* + \operatorname{vec} \sum_{k=1}^{K} G_{1k}^{*\prime} H_{t-1} G_{1k}^* \\ &= \operatorname{vec}(C_0^{*\prime} C_0^*) + \sum_{k=1}^{K} (A_{1k}^* \otimes A_{1k}^*)' \operatorname{vec}(\epsilon_{t-1} \epsilon_{t-1}') + \sum_{k=1}^{K} (G_{1k}^* \otimes G_{1k}^*)' \operatorname{vec}(H_{t-1}) \\ &= (C_0^* \otimes C_0^*)' \operatorname{vec}(I_n) + \sum_{k=1}^{K} (A_{1k}^* \otimes A_{1k}^*)' \eta_{t-1} + \sum_{k=1}^{K} (G_{1k}^* \otimes G_{1k}^*)' h_{t-1}. \end{aligned} \tag{A.3}$$

Now if relations (2.5) hold, then (A.3) becomes

$$h_t = C_0 + A_1 \eta_{t-1} + G_1 h_{t-1},$$

which is (A.1), proving sufficiency. Necessity can be shown by noting that (A.1) and (A.3) must hold for all ϵ_{t-1}, so by appropriate choice of ϵ_{t-1}, each column of A_1 can be equated individually with each column of $\sum_{k=1}^{K} (A_{1k}^* \otimes A_{1k}^*)'$. For example, letting $\epsilon_{t-1}' = (1,0,\ldots,0)$ establishes equality of the first column of A_1 with the first column of $\sum_{k=1}^{K} (A_{1k}^* \otimes A_{1k}^*)'$. Necessity of the rest of relations (2.5) can be shown in the same way. ■

Proof for Proposition 2.5. For simplicity, we look at GARCH(1,1) models, but the generalization to GARCH(p,q) models is obvious. The *BEKK* parameterization is

$$H_t = C_0^{*\prime} C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime} \epsilon_{t-1} \epsilon_{t-1}' A_{1k}^* + \sum_{k=1}^{K} G_{1k}^{*\prime} H_{t-1} G_{1k}^*.$$

The proof proceeds by induction. First, H_t is positive definite for $t = 1$: The term $\sum_{k=1}^{K} A_{1k}^{*\prime} \epsilon_0 \epsilon_0' A_{1k}^*$ is positive semidefinite because $\epsilon_0 \epsilon_0'$ is positive semidefinite. Also, if the null space condition holds, then

$$C_0^{*\prime} C_0^* + \sum_{k=1}^{K} G_{1k}^{*\prime} H_0 G_{1k}^* \tag{A.4}$$

is positive definite. This is clearly true if C_0^* or any G_{1k}^* has full rank, but to show that the null space condition is sufficient, note that expression (A.4) is positive definite if and only if

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$$x'\left(C_0^{*\prime}C_0^* + \sum_{k=1}^{K} G_{1k}^{*\prime}H_0G_{1k}^*\right)x > 0 \qquad \forall x \neq 0$$

or

$$(C_0^*x)'(C_0^*x) + \sum_{k=1}^{K} (H_0^{1/2}G_{ik}^*x)'(H_0^{1/2}G_{ik}^*x) > 0 \qquad \forall x \neq 0, \tag{A.5}$$

where $H_0 = H_0^{1/2\prime}H_0^{1/2}$ and $H_0^{1/2}$ is full rank. But defining $N[P]$ to be the null space of the matrix P, (A.5) is true if and only if

$$N[C_0^*] \cap N[H_0^{1/2}G_{11}^*] \cap \cdots \cap N[H_0^{1/2}G_{1K}^*] = \varnothing.$$

Noting that $N[H_0^{1/2}G_{1k}^*]$ is the same as $N[G_{1k}^*]$ because $H_0^{1/2}$ is full rank gives the desired result—(A.4) is positive definite if and only if $N[C_0^*] \cap N[G_{11}^*] \cap \cdots \cap N[G_{1K}^*] = \varnothing$.

Now suppose that the statement is true for $n = \tau$; i.e., suppose

$$H_\tau = C_0^{*\prime}C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime}\epsilon_{\tau-1}\epsilon_{\tau-1}'A_{1k}^* + \sum_{k=1}^{K} G_{1k}^{*\prime}H_{\tau-1}G_{1k}^* \tag{A.6}$$

is positive definite. Then,

$$H_{\tau+1} = C_0^{*\prime}C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime}\epsilon_\tau\epsilon_\tau'A_{1k}^* + \sum_{k=1}^{K} G_{1k}^{*\prime}H_\tau G_{1k}^*$$

is positive definite. First, following a similar line of argument as above, the term $\sum_{k=1}^{K} A_{1k}^{*\prime}\epsilon_\tau\epsilon_\tau'A_{1k}^*$ is positive semidefinite. Second, the term

$$C_0^{*\prime}C_0^* + \sum_{k=1}^{K} G_{1k}^{*\prime}H_\tau G_{1k}^*$$

is positive definite if and only if the null space condition holds, because H_τ is positive definite by the induction assumption. This can be shown by following the identical steps as in the preceding paragraph, replacing H_0 with H_τ. So

$$H_{\tau+1} = C_0^{*\prime}C_0^* + \sum_{k=1}^{K} A_{1k}^{*\prime}\epsilon_\tau\epsilon_\tau'A_{1k}^* + \sum_{k=1}^{K} G_{1k}^{*\prime}H_\tau G_{1k}^*$$

is positive definite, meaning H_τ is positive definite for all τ. ■

Proof for Proposition 2.6. First, suppose $C_0 = \text{vec}(\Omega)$, where Ω is positive definite. Then, $\Omega = C_0^{*\prime}C_0^*$ for some triangular C_0^*. Therefore, $C_0 = \text{vec}(\Omega) = \text{vec}(C_0^{*\prime}C_0^*)$.

To prove the remaining equations, we again assume $p = q = 1$. The extension to higher order GARCH models is obvious. Also, we only show that the equations relating A_i and A_{ik}^* hold; the proof for the equations relating G_i and G_{ik}^* is directly analogous. First, consider the second term in the *diagonal vec* model, which we will label $\tilde{H}_t$. This term can be written as

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$$\tilde{H}_t = \begin{bmatrix} \tilde{a}_{11}\epsilon_{1,t-1}^2 & \tilde{a}_{12}\epsilon_{1,t-1}\epsilon_{2,t-1} & \cdots & \tilde{a}_{1n}\epsilon_{1,t-1}\epsilon_{n,t-1} \\ \tilde{a}_{21}\epsilon_{2,t-1}\epsilon_{1,t-1} & \tilde{a}_{22}\epsilon_{2,t-1}^2 & \cdots & \tilde{a}_{2n}\epsilon_{2,t-1}\epsilon_{n,t-1} \\ \vdots & \vdots & \cdots & \vdots \\ \tilde{a}_{n1}\epsilon_{n,t-1}\epsilon_{1,t-1} & \tilde{a}_{n2}\epsilon_{n,t-1}\epsilon_{2,t-1} & \cdots & \tilde{a}_{nn}\epsilon_{n,t-1}^2 \end{bmatrix}. \tag{A.7}$$

If H_t is positive definite for all realizations of ϵ_t, then $\tilde{H}_t$ must be positive semidefinite for all realizations of ϵ_t. But $\tilde{H}_t$ is positive semidefinite if and only if all its principal minors are nonnegative. Also, the principal minors of (A.7) have the same sign as the principal minors of the matrix

$$\tilde{A} = \begin{bmatrix} \tilde{a}_{11} & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} \\ \tilde{a}_{21} & & & \tilde{a}_{2n} \\ \vdots & & & \vdots \\ \tilde{a}_{n1} & & \cdots & \tilde{a}_{nn} \end{bmatrix},$$

implying that if $\tilde{H}_t$ is positive semidefinite for all realizations of ϵ_t, then $\tilde{A}$ must be positive semidefinite. But if $\tilde{A}$ is positive semidefinite, then it can always be decomposed into $\tilde{A} = B'B$ with B triangular. Define B to be

$$B = \begin{bmatrix} b_{11,n} & b_{11,n-1} & \cdots & b_{11,1} \\ 0 & b_{22,n-1} & \cdots & b_{22,1} \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & b_{nn,1} \end{bmatrix}.$$

Then,

$$\tilde{A} = B'B = \begin{bmatrix} \sum_{k=1}^{n} b_{11,k}^2 & \sum_{k=1}^{n-1} b_{11,k}b_{22,k} & \cdots & b_{11,1}b_{nn,1} \\ \cdot & \sum_{k=1}^{n-1} b_{22,k}^2 & \cdots & b_{22,1}b_{nn,1} \\ \vdots & & & \vdots \\ \cdot & & & b_{nn,1}^2 \end{bmatrix}.$$

So if H_t is positive definite, then there exists $b_{ij,k}$ such that

$$\tilde{H}_t = \begin{bmatrix} \sum_{k=1}^{n} b_{11,k}^2\epsilon_{1,t-1}^2 & \sum_{k=1}^{n-1} b_{11,k}b_{22,k}\epsilon_{1,t-1}\epsilon_{2,t-1} & \cdots & b_{11,1}b_{nn,1}\epsilon_{1,t-1}\epsilon_{n,t-1} \\ \cdot & \sum_{k=1}^{n-1} b_{22,k}^2\epsilon_{2,t-1}^2 & \cdots & b_{22,1}b_{nn,1}\epsilon_{2,t-1}\epsilon_{n,t-1} \\ \vdots & & & \vdots \\ \cdot & & & b_{nn,1}^2\epsilon_{n,t-1}^2 \end{bmatrix}. \tag{A.8}$$

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But notice that in the *diagonal BEKK* model, if we define

$$A_{ik}^{*} = \begin{bmatrix} a_{11,k}^{*} & & & & & & \\ & a_{22,k}^{*} & & & 0 & & \\ & & \ddots & & & & \\ & & & a_{rr,k}^{*} & & & \\ & & & & 0 & & \\ & 0 & & & & \ddots & \\ & & & & & & 0 \end{bmatrix},$$

where $r = n - k + 1$, i.e., if we define $A_{1k}^{*} = \text{diag}\{a_{11,k}^{*}, a_{22,k}^{*}, \ldots, a_{rr,k}^{*}, 0, \ldots, 0\}$, then evaluation of $H_t = \sum_{k=1}^{n} A_{1k}^{*\prime} \epsilon_{t-1} \epsilon_{t-1}' A_{1k}^{*}$ gives

$$H_t = \begin{bmatrix} \sum_{k=1}^{n} a_{11,k}^{*2} \epsilon_{1,t-1}^{2} & \sum_{k=1}^{n-1} a_{11,k}^{*} a_{22,k}^{*} \epsilon_{1,t-1} \epsilon_{2,t-1} & \cdots & a_{11,1}^{*} a_{nn,1}^{*} \epsilon_{1,t-1} \epsilon_{n,t-1} \\ \cdot & \sum_{k=1}^{n-1} a_{22,k}^{*2} \epsilon_{2,t-1}^{2} & \cdots & a_{22,1}^{*} a_{nn,1}^{*} \epsilon_{2,t-1} \epsilon_{n,t-1} \\ \vdots & & & \vdots \\ \cdot & & & a_{nn,1}^{*2} \epsilon_{n,t-1}^{2} \end{bmatrix}.$$

Comparing this with (A.8), we see that if H_t is positive definite then we can always choose each $a_{ii,k}^{*}$ in the diagonal A_{1k}^{*} matrices to be equal to $b_{ii,k}$ from the decomposition of $\tilde{A}$, implying that if H_t is positive definite then it can always be written in the *BEKK* framework. ■

Proof for Proposition 2.7. Again, we show this for a GARCH(1,1) model, but the extension to GARCH(p,q) models is trivial. Also, we prove this only for the *vec* model. The proof for the *BEKK* model can be obtained by substituting relations (2.5) into this proof. Let E_{t-1} be the expectations operator, conditioned on the information set $\Im_{t-1}$. Then,

$$E_{t-1}\eta_t = \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1 \eta_{t-i}]$$

$$\begin{aligned} E_{t-2}\eta_t &= E_{t-2}E_{t-1}\eta_t \\ &= E_{t-2}\sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1 \eta_{t-i}] \\ &= \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1 E_{t-2}\eta_{t-i}] \\ &= C_0 + A_1 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1 \eta_{t-i-1}] + \sum_{i=2}^{\infty} G_1^{i-1}[C_0 + A_1 \eta_{t-i}] \end{aligned}$$

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$$= C_0 + A_1 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-1}] + G_1 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-1}]$$

$$= C_0 + [A_1 + G_1] \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-1}]$$

$$E_{t-3}\eta_t = E_{t-3}E_{t-2}\eta_t$$

$$= C_0 + [A_1 + G_1] \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1 E_{t-3}\eta_{t-i-1}]$$

$$= C_0 + [A_1 + G_1](C_0 + A_1 h_{t-2}) + [A_1 + G_1] \sum_{i=2}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-1}]$$

$$= C_0 + (A_1 + G_1)C_0 + (A_1 + G_1)A_1 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-2}]$$

$$+ (A_1 + G_1) \sum_{i=2}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-1}]$$

$$= C_0 + (A_1 + G_1)C_0 + (A_1 + G_1)A_1 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-2}]$$

$$+ (A_1 + G_1)G_1 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-2}]$$

$$= C_0 + (A_1 + G_1)C_0 + (A_1 + G_1)^2 \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-2}]$$

$$\vdots$$

$$E_{t-\tau}\eta_t = [I + (A_1 + G_1) + \cdots + (A_1 + G_1)^{\tau-2}]C_0$$

$$+ (A_1 + G_1)^{\tau-1} \sum_{i=1}^{\infty} G_1^{i-1}[C_0 + A_1\eta_{t-i-\tau+1}].$$

It is widely known that for any square matrix Z, $Z^\tau \to 0$ as $\tau \to \infty$ if and only if all the eigenvalues of Z are less than one in modulus and that the eigenvalues of Z are less than one in modulus if and only if $[I + Z + Z^2 + \cdots] \to (I - Z)^{-1}$. Therefore, $E_{t-\tau}\eta_t$ converges in probability (as $\tau \to \infty$) to $[I - A_1 - G_1]^{-1}C_0$ if and only if the eigenvalues of $(A_1 + G_1)$ are all less than one in modulus. Also, by the law of iterated expectations, $E(\epsilon_t\epsilon'_{t+\gamma}) = E[E_t(\epsilon_t\epsilon_{t+\gamma})] = 0$ for all $\gamma \neq 0$. Therefore, $E(\epsilon_t\epsilon_{t+\gamma})$ exists and depends only on γ for all integers t. ■

Proof for Proposition 3.1. Again, the proof is presented for the *vec* model, but an application of relations (2.5) reveals that the proof also applies for the *BEKK* model. Suppose

$$\epsilon_t | \Im_{t-1} \sim N(0, H_t)$$

$$h_t = C_0 + \sum_{i=1}^{q} A_i\eta_{t-i} + \sum_{i=1}^{p} G_i h_{t-i}.$$

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Define $\epsilon_t^* \equiv P\epsilon_t$, $h_t^* \equiv \text{vec}\, PH_tP'$ and $\Psi \equiv (P \otimes P)$. Then, ignoring the summation limits, $\epsilon_t^* \mid \Im_{t-1} \sim N(0, PH_tP')$ with

$$
\begin{aligned}
h_t^* &= \text{vec}\, PH_tP' \\
&= (P \otimes P)\text{vec}\, H_t \\
&= \Psi h_t \\
&= \Psi C_0 + \Psi \sum A_i \eta_{t-i} + \Psi \sum G_i h_{t-i} \\
&= \Psi C_0 + \Psi \sum A_i \,\text{vec}(\epsilon_{t-i}\epsilon_{t-i}') + \Psi \sum G_i \,\text{vec}\, H_{t-i} \\
&= \Psi C_0 + \Psi \sum A_i \,\text{vec}(P^{-1}P\epsilon_{t-i}\epsilon_{t-i}'P'P^{-1\prime}) + \Psi \sum G_i \,\text{vec}(P^{-1}PH_{t-i}P'P^{-1\prime}) \\
&= \Psi C_0 + \Psi \sum A_i (P^{-1} \otimes P^{-1})\text{vec}(P\epsilon_{t-i}\epsilon_{t-i}'P') \\
&\quad + \Psi \sum G_i (P^{-1} \otimes P^{-1})\text{vec}(PH_{t-i}P') \\
&= \Psi C_0 + \Psi \sum A_i (P^{-1} \otimes P^{-1})\text{vec}(\epsilon_{t-i}^*\epsilon_{t-i}^{*\prime}) + \Psi \sum G_i (P^{-1} \otimes P^{-1}) h_{t-i}^* \\
&= \Psi C_0 + \Psi \sum A_i (P^{-1} \otimes P^{-1}) \eta_{t-i}^* + \Psi \sum G_i (P^{-1} \otimes P^{-1}) h_{t-i}^* \\
&= \Psi C_0 + \sum \Psi A_i \Psi^{-1} \eta_{t-i}^* + \sum \Psi G_i \Psi^{-1} h_{t-i}^*.
\end{aligned}
$$

Hence, $P\epsilon_t$ follows a GARCH process of the same order as ϵ_t. ■

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